

# Aryan Jain

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## EDUCATION

### University of California, Berkeley

*Bachelor of Arts in Data Science*

GPA: 4.00

Aug. 2025 – May 2029

## EXPERIENCE

### Chipchop

*Software Engineering Intern*

Jan. 2026 – Present

Berkeley, CA

- Built backend features for an FPGA debugging platform using Python, FastAPI, and WebSockets, enabling a multi-agent system to analyze and troubleshoot Verilog/VHDL projects in a Vivado-style environment.
- Improved developer workflows in the React + TypeScript IDE by enhancing evaluation and debugging tooling to surface agent outputs, run artifacts, and project context directly within the IDE interface and debugging panels.
- Strengthened pipeline reliability by adding validation, diagnostics-based error handling, and automated tests with pytest and unittest.mock, stabilizing multi-agent execution on larger HDL project evaluations and debugging runs.

### Undergraduate Lab at Berkeley – Astronomy Division

*Researcher*

September 2025 – Present

Berkeley, CA

- Exploring protoplanetary disk chemistry to predict if forming exoplanets will be suitable for nurturing life.
- Use CARTA, CASSIS, and Python to model spectra and make predictions with data extracted from datacubes.
- Present a poster of the group's research and findings at a symposium to graduate students and professors.

### Quantum Materials Research Group

*Researcher*

Jul. 2024 – Dec. 2024

Rochester, NY

- Conducted database screening with Python on 200k+ materials to identify more efficient PV cell materials.
- Worked under Prof. Sobhit Singh to optimize sunlight capture and reduce carbon emissions in the energy sector.
- Presented results to 20+ graduate researchers in a Materials Science seminar at the University of Rochester.

## PROJECTS

### Multi-Agent Research Assistant

Feb. 2026

- Engineered a multi-agent AI system that retrieves academic papers, extracts structured evidence, critiques study quality, and synthesizes citation-grounded conclusions through a role-specialized agent pipeline.
- Designed a FastAPI backend orchestrator coordinating Planner, Retriever, Extractor, Critic, Synthesizer, and Referee agents with schema-enforced outputs and verification loops to mitigate model hallucinations.
- Built a React + TypeScript interface with local conversation persistence, collapsible evidence panels, and live progress updates to expose intermediate agent reasoning and improve system transparency.

### Agentic Calendar Assistant – CalHacks 12.0

Oct. 2025

- Built an agentic AI calendar assistant that automates scheduling and task management, providing context-aware recommendations such as gym timing, meal suggestions based on user preference, and dynamic to-do list updates.
- Integrated retrieval-augmented generation and Lava LLM API to support natural, chat-based event planning.
- Designed with Python, FastAPI, LangChain, and React to create a powerful and intuitive application.

## VOLUNTEER & LEADERSHIP EXPERIENCE

### STEAM Foundation Coding Camp

Feb. 2022 – May 2025

- Taught Python to 100+ beginner students and developed/managed instructional materials through Google Classroom.

### VEX Robotics Team (9971A) Co-Founder

Jun. 2023 – Apr. 2025

- Recruited members and secured thousands in funding for a new, private VEX robotics team founded in my garage.
- Designed, built, and programmed robots for VRC tournaments, qualifying for the state-level twice over two seasons.

## AWARDS & SKILLS

**Awards:** Gold PVSA (250+ hrs), 6th/1800+ in Project Empower Hackathon for creating a scholarship probability model, 3rd in Dublin Hackathon, Gold in National Spanish Exam

**Programming:** Python, Java, JS, HTML/CSS, TensorFlow, PyTorch, NumPy, FastAPI, LangChain, React, SQL